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Valve for Supplying and Purging Lubricants from Systems

Abstract

A valve includes a housing and a plunger. The housing defines a chamber, a set of ports, and a set of conduits. The plunger is disposed within the chamber and is configured to be moved to an actuated state in which the set of ports are fluidly coupled to each other through a first section of the chamber to define a first passageway and the set of conduits are fluidly coupled to each other through a second section of the chamber to define a second passageway. In the actuated state, the valve facilitates an inflow of a lubricant into a system through the first passageway to cause a concomitant purge and release of a used lubricant from the system through the second passageway.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to a valve for facilitating lubricant transfer with respect to a system. More particularly, the present disclosure relates to a valve for supplying a new lubricant to a system and, in process, purging a used lubricant from the system.

BACKGROUND

[0002] Applications, such as, mining, earth moving, and construction, typically involve the use of machines, such as excavators, shovels, loaders, articulated trucks, and the like, to carry out one or more tasks at a worksite. Such machines generally include systems in which components interact and/or move with respect to one another so that they can perform their intended function(s) effectively. As an example, articulation systems in hitch systems of an articulated truck includes a myriad of components that are configured to interact and/or move with respect to one other to facilitate a turning of a forward portion of a frame of the articulated truck with respect to the rearward portion of the frame of the articulated truck, during machine travel.

[0003] Generally, such components in such systems are supplied with one or more lubricants (e.g., grease) so that they can interact and/or move with respect to each other with relative ease. However, after a period of use, such lubricants may deteriorate in quality and/or lose one or more of their properties (e.g., viscosity), and may accordingly need a replacement. Replacing a lubricant (e.g., a used lubricant) or a lubricant replacement procedure generally involves discarding a used lubricant from the system and replenishing the system with a new lubricant. An improperly discarded lubricant may contaminate the environment. Moreover, if the system to be replaced with the new lubricant is a sealed system (e.g., sealed to prevent the components and/or the lubricant running therein from external contamination), a difficulty associated with the lubricant replacement procedure is generally further compounded.

SUMMARY

[0004] In one aspect, the disclosure is directed to a valve. The valve includes a housing and a plunger. The housing defines a chamber, a set of ports, and a set of conduits. The plunger is disposed within the chamber and is configured to be moved to an actuated state in which the set of ports are fluidly coupled to each other through a first section of the chamber to define a first passageway and the set of conduits are fluidly coupled to each other through a second section of the chamber to define a second passageway. In the actuated state, the valve facilitates an inflow of a lubricant into a system through the first passageway to cause a concomitant purge and release of a used lubricant from the system through the second passageway.

[0005] In another aspect, the disclosure relates to a method for replenishing a lubricant in a system. The method includes providing a valve. The valve includes a housing and a plunger. The housing defines a chamber, a set of ports, and a set of conduits. The plunger is disposed within the chamber and is configured to be moved to an actuated state in which the set of ports are fluidly coupled to each other through a first section of the chamber to define a first passageway and the set of conduits are fluidly coupled to each other through a second section of the chamber to define a second passageway. The method further includes fluidly coupling the first passageway to a lubricant supply line to supply lubricant to the system and fluidly coupling the second passageway

to a lubricant release line to release lubricant from the system. In the actuated state, the valve facilitates an inflow of a lubricant into the system through the first passageway to cause a concomitant purge and release of a used lubricant from the system through the second passageway. [0006] Machines, such as excavators, shovels, loaders, articulated trucks, and the like, generally include systems to carry out one or more tasks. Such systems include multiple components that interact and/or move with respect to one another so that they can perform their intended function(s) effectively, and thus, such systems need to be properly lubricated and/or greased. A lubricant (e.g., grease) applied in such systems may be replaced after a period of use. If a good seal exists in such systems, it is generally difficult to replenish the system with a new lubricant and purge the system of an old lubricant. Moreover, an improper purging of a used lubricant may also contaminate the environment.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 is a side view of an exemplary machine having a forward frame portion, a rearward frame portion, and a hitch system that pivotally couples the forward frame portion to the rearward frame portion, in accordance with an embodiment of the present disclosure;

[0008] FIG. 2 is an enlarged view of the hitch system illustrated in conjunction with a valve that facilitates lubricant transfer with respect to the hitch system, in accordance with an embodiment of the present disclosure;

[0009] FIG. 3 is the enlarged view of the hitch system illustrated with the valve, in accordance with another embodiment of the present disclosure; and

[0010] FIGS. 4 and 5 are various views illustrating different working or actuation states of the valve, in accordance with an embodiment of the present disclosure.

DETAILED DESCRIPTION

[0011] Reference will now be made in detail to specific embodiments or features, examples of which are illustrated in the accompanying drawings. Generally, corresponding reference numbers may be used throughout the drawings to refer to the same or corresponding parts, e.g., **1**, **1'**, **1''**, **101** and **201** could refer to one or more comparable components used in the same and/or different depicted embodiments.

[0012] Referring to FIG. 1, an exemplary machine (i.e., machine **100**) is shown. The machine **100** may include an articulated machine **104** (e.g., an articulated dump truck **104'**) having a dump body **108** within which a payload (not shown) may be received. The payload may be transferred or hauled from one location to another location at a worksite **112** by the machine **100**. The payload may include, but not limited to, rocks, ores, minerals, and/or other disintegrated particles generally found at a worksite. Examples of the worksite **112** may include a construction site, a mine site, a landfill, a quarry, an underground mine site, and the like. Further, locations of the worksite **112** between which the machine **100** may traverse back and forth during operations may include one or more load locations (not shown) from where the machine **100** may receive the material, and one or more dump locations (not shown) up to where the machine **100** may traverse to so as to dump or release the material from the dump body **108**.

[0013] The machine **100** may include a frame **116**, an operator cab **120**, and one or more traction devices **124** supported on the frame **116**. The machine **100** may also include several other parts and sub-systems, such as a power compartment **128** that may house a power source (e.g., an internal combustion engine) (not shown) for powering several functions of the machine **100**, including, but not limited to, providing power to the traction devices **124** such that the machine **100** may be propelled or moved over a ground surface **132** of the worksite **112**. The traction devices **124** may include wheels or endless crawler tracks, or a combination thereof.

[0014] The frame **116** may include a rigid structure to which several other components (and/or sub-components) of the machine **100** may be coupled to. If the machine **100** were to include the articulated machine **104**, such as the articulated dump truck **104'**, the frame **116** may include a split frame configuration that defines a forward frame portion **136** and a rearward frame portion **140**. The forward frame portion **136** may be pivotably coupled to the rearward frame portion **140** by way of a hitch system **144** disposed at a hitch location **148** (also see FIG. 2). As an example, the hitch system **144** may be disposed in between the forward frame portion **136** and the rearward frame portion **140**). By way of the hitch system **144**, the forward frame portion **136** may pivot or articulate relative to the rearward frame portion **140**, during machine movement, facilitating a steering of the machine **100** during machine movement.

[0015] Referring to FIGS. 1 and 2, for enabling a pivotal coupling between the forward frame portion **136** and the rearward frame portion **140**, portions of the hitch system **144** may be respectively accommodated within corresponding receiving portions or receptacles (e.g., a forward receptacle **152** and a rearward receptacle **156**) respectively structured and arranged in the forward frame portion **136** and the rearward frame portion **140**. Also, galleries, e.g., a first gallery **154** and a second gallery **158** may be defined in the forward frame portion **136**. The first gallery **154** and the second gallery **158**, when applied in concert or combination, may allow a lubricant transfer (e.g., a lubricant supply and a lubricant purge/release) to be carried out with respect to one or more components or sub-systems of the hitch system **144** such that a proper working of the hitch system **144** may be attained. As an example, the first gallery **154** may facilitate a lubricant supply into such systems and the second gallery **158** may facilitate a lubricant release or a lubricant purge from such systems. The forward frame portion **136** may also define a duct **166**, as shown. Additional details related to each of the first gallery **154**, second gallery **158**, and the duct **166**, shall be set out later in the present disclosure.

[0016] Further, and as an example, the forward frame portion **136** may support the operator cab **120** that, apart from housing various parts and systems, such as controls, etc., of the machine **100**, may also accommodate/station one or more operators (not shown) of the machine **100** for the control of the many functions of the machine **100**. The forward frame portion **136** may also support the power compartment **128** housing the power source (e.g., the internal combustion engine) that may power the traction devices **124** of the machine **100**. The rearward frame portion **140** may support the dump body **108** of the machine **100**.

[0017] During a straight line travel of the machine **100**, the forward frame portion **136** may be aligned (i.e., to be in line) with respect to the rearward frame portion **140**. However, during a movement of the machine **100** either towards a left of the machine **100** or towards a right of the machine **100**, the forward frame portion **136** of the machine **100** may articulate or be pivoted relative to the rearward frame portion **140** of the machine **100**, as facilitated by way of the hitch system **144**.

[0018] It may be noted that the terms 'forward' and 'rearward', as used herein, are in relation to an exemplary direction of travel of the machine **100**, as represented by arrow, T, in FIG. 1. Said direction of travel, T, may be the direction in which the machine **100** may generally travel so as to move or shuttle between a load location and a dump location of the worksite **112**. Said direction of travel, T, may be exemplarily defined from a rearward end **160** of the machine **100** towards a forward end **164** of the machine **100**. Also, the terms 'left' and 'right' as used in the present disclosure may be understood when viewing the machine **100** from the rearward end **160** towards the forward end **164**.

[0019] One or more aspects of the present disclosure are discussed in connection with a system of the machine **100** that includes components interacting or moving with respect to one other, with such systems using lubricants, e.g., grease, so that they can uninterruptedly move with respect to one another to perform an intended task. As an example, said system includes or corresponds to the hitch system **144** and/or to one or more sub-systems of the hitch system **144**. It is possible however

for those aspects to be applied to multiple other systems or sub-systems of the machine **100**. Moreover, it is also possible that aspects of the present disclosure be applicable to various systems of various other machines as well, such as excavators, loaders, graders, off-highway trucks, dozers, draglines, which use lubricants for a smooth and/or an uninterrupted interaction and/or movement between any two or more of their components.

[0020] Use of one or more aspects of the present disclosure also may be extended to systems applied in applications, such as marine applications, gas turbine applications, or to stationary machine applications (e.g., generator sets), and in which a set of components may require a lubricant (e.g., grease) such that they can smoothly interact and/or move with respect to each other. Effectively, references to the hitch system **144** and/or its sub-systems, and/or the machine **100** itself, as may be viewed and understood from the forthcoming disclosure below, are to be viewed as being purely exemplary.

[0021] The hitch system **144** includes a number of components that interact with each other to perform the intended function of facilitating and/or enabling the forward frame portion **136** to articulate or pivot with respect to the rearward frame portion **140**. To this end, the hitch system **144** includes a hitch pin **168** and one or more sub-systems, such as a first bearing system **172** and a second bearing system **176**. Further, the hitch system **144** may also include a lubricant case **180** and a seal **184**. The hitch system **144** may include various other components, but for simplicity, they have not been discussed as they may be contemplated and applied by those with skill in the art.

[0022] The first bearing system **172** may include a first inner race **188** and a first outer race **192**. The first inner race **188** and the first outer race **192** may be slidably disposed with respect to each other. As an example, the first outer race **192** may be fixedly retained (e.g., by press fitting and/or by bolting) against and/or into the forward receptacle **152** of the forward frame portion **136** such that the first outer race **192** may move along with a movement (e.g., rotary movement) of the forward frame portion **136** with respect to the first inner race **188** as the forward frame portion **136** moves (e.g., pivots or articulates) relative to the rearward frame portion **140**. Further, and as an example, the first inner race **188** may be fixedly retained (e.g., by press fitting and/or bolting) with respect to a portion of the hitch pin **168**.

[0023] The first outer race **192** may define a lubricant channel (e.g., a first lubricant channel **196**) through which a lubricant (e.g., grease) may be introduced into an interface (e.g., a first interface **200**) defined between the first inner race **188** and the first outer race **192**. In so doing, a movement (e.g., a slidable and rotary movement) between the first inner race **188** and the first outer race **192** may be achieved with relative ease, and which in turn allows the forward frame portion **136** to move (e.g., pivot or articulate) relative to the hitch pin **168** (and thus relative to the rearward frame portion **140**). In some embodiments, a lubricant, such as grease, may be supplied into the first lubricant channel **196** through the first gallery **154** of the forward frame portion **136**. Combinedly, the first lubricant channel **196** and the first gallery **154** may function as a lubricant supply line **198** for supply of a lubricant (e.g., a new lubricant) to the first bearing system **172**.

[0024] Similar to the configuration of the first bearing system **172**, the second bearing system **176** may include a second inner race **204** and a second outer race **208**. The second inner race **204** and the second outer race **208** may be slidably disposed with respect to each other. As an example, the second outer race **208** may be fixedly retained (e.g., by press fitting and/or by bolting) against and/or into the rearward receptacle **156** of the rearward frame portion **140** such that the second outer race **208** may move along with a movement (e.g., rotary movement) of the rearward frame portion **140** with respect to the second inner race **204** as the rearward frame portion **140** moves (e.g., pivots or articulates) relative to the forward frame portion **136**. Further, and as an example, the second inner race **204** may be fixedly retained (e.g., by press fitting and/or bolting) to the hitch pin **168**.

[0025] The second outer race **208** may also define a lubricant channel (e.g., a second lubricant channel **212**) through which a lubricant (e.g., grease) may be introduced into an interface (e.g.,

second interface **216**) defined between the second inner race **204** and the second outer race **208**. In so doing, a movement (e.g., a slidable and rotary movement) between the second inner race **204** and the second outer race **208** may be achieved with relative ease, in turn allowing the rearward frame portion **140** to move (e.g., pivot or articulate) relative to the hitch pin **168** (and thus relative to the forward frame portion **136**).

[0026] The lubricant case **180** may be configured to collect a volume of lubricant (e.g., new lubricant) exiting the interface (e.g., the first interface **200**) defined between the first inner race **188** and the first outer race **192**. The lubricant case **180** may be positioned under the first bearing system **172** and a collection of the lubricant (e.g., new lubricant) from the first interface **200** may be attained under the action of gravity or because of an inflow of the lubricant caused from a surplus lubricant pressure developed at the interface, e.g., during a replenishment (or a supply) of the lubricant (e.g., a new lubricant) at or into the first interface **200** through the first lubricant channel **196**. Further, the lubricant case **180** may define an outlet **214** to direct a lubricant out from the lubricant case **180** into the duct **166** formed in the forward frame portion **136**.

[0027] The seal **184** may be provided or disposed above the first bearing system **172**, as shown. The seal **184** may function to shield the first bearing system **172** and the lubricant housed within or around the first bearing system **172** from an outside environment. The seal **184** may also contain or restrict the lubricant within a boundary defined by or around the first bearing system **172**. The seal **184** may define an internal seal chamber **220** to accommodate the many components of the first bearing system **172**, such as portions of the hitch pin **168**, and to also receive a volume of lubricant flowing out from the lubricant case **180** through the duct **166**.

[0028] Referring to FIGS. **2** to **5**, a valve **230** for a facilitating a lubricant transfer with respect to the first bearing system **172** of the hitch system **144** is described. The expression 'lubricant transfer', as noted above or as used in the present disclosure, may mean or involve both a supply of a new lubricant into the first bearing system **172** or into the first interface **200** of the first bearing system **172** and a simultaneous or concomitant release or purge of a used lubricant from the first bearing system **172**. In some embodiments, the lubricant transfer may correspond to a controlled lubricant transfer. In other words, the lubricant transfer may involve a metered supply of an amount of a new lubricant into the first bearing system **172** or the first interface **200** of the first bearing system **172** and an equitable release or purge of an amount of the used lubricant from the first bearing system **172** or the first interface **200** of the first bearing system **172** resulting from said metered supply of the new lubricant. The valve **230** includes a housing **234**, a plunger **238**, a biasing member **242**, and a plug **246**.

[0029] Referring to FIGS. **4** and **5**, the housing **234** may include a generally elongated cylindrical structure defining a primary axial end **250** and a secondary axial end **254**. The primary axial end **250** and the secondary axial end **254** may be disposed axially oppositely to each other. The housing **234** also defines a chamber **258** which may extend between the primary axial end **250** and the secondary axial end **254**. The chamber **258** may impart a generally hollow cylindrical profile to the cylindrical structure of the housing **234**. Other shapes or profiles of the housing **234** may be contemplated. Further, the chamber **258** may define a chamber axis **262** passing between and/or through the primary axial end **250** and the secondary axial end **254**.

[0030] A wall **264** of the housing **234** may (at least partially) surround the chamber **258**. As an example, a portion (e.g., a first portion **268**) of the wall **264** may be present at the primary axial end **250** of the housing **234** to close the chamber **258** at the primary axial end **250** of the housing **234** while another portion (e.g., a second portion **272**) of the wall **264** may be present annularly around the chamber **258** (e.g., which may run annularly around the chamber axis **262**). A portion of the wall **264** may however be omitted at the secondary axial end **254** of the housing **234**. In that manner, the chamber **258** may be generally closed or blind to the primary axial end **250** of the housing **234** and open to the secondary axial end **254** of the housing **234**. Further, the chamber **258** may also define an opening **276** at the secondary axial end **254** for gaining access into the chamber

258 from the secondary axial end **254**.

[0031] The housing **234** may also define a set of ports **280** and a set of conduits **284**. The set of ports **280** may correspond to a first port **280'** and a second port **280''**. A higher number of the set of ports **280** may be contemplated. As an example, the first port **280'** may be formed or disposed at the primary axial end **250** of the housing **234** and may extend through a thickness, **T1**, of the first portion **268** of the wall **264** such that the first port **280'** may facilitate fluid communication between the chamber **258** and an environment external to the housing **234**. The first port **280'** may define a first port axis **288**, which may be co-axial or parallel to the chamber axis **262**. The expression 'generally closed or blind', as used above, with respect to the chamber **258** means that the first port **280'** may be closed to the chamber **258** in one or more working states of the valve **230**.

[0032] As an example, the second port **280''** may be formed or disposed on the second portion **272** of the wall **264** of the housing **234** and may extend through a thickness, **T2**, of the second portion **272** of the wall **264** such that the second port **280''** may facilitate fluid communication between the chamber **258** and the environment external to the housing **234**. The second port **280''** may define a second port axis **292**. In some embodiments, the second port axis **292** may be disposed orthogonally to the first port axis **288**.

[0033] The set of conduits **284** may correspond to a first conduit **284'** and a second conduit **284''**. A higher number of the set of conduits **284** may be contemplated. Both the first conduit **284'** and the second conduit **284''** may be formed or disposed on the second portion **272** of the wall **264** of the housing **234** and may each independently extend through the thickness, **T2**, of the second portion **272** of the wall **264** such that each of first conduit **284'** and the second conduit **284''** may facilitate fluid communication between the chamber **258** and the environment external to the housing **234**. Although not limited, the first conduit **284'** and the second conduit **284''** may be disposed diametrically oppositely to each other. Further, the first conduit **284'** may define a first conduit axis **296** and the second conduit **284''** may define a second conduit axis **300**. In some embodiments, the first conduit axis **296** and the second conduit axis **300** may be co-axially aligned to each other or may be parallel to each other. Further, it may be noted that the set of ports **280** may be positioned towards (or relatively closer to) the primary axial end **250** of the housing **234** while the set of conduits **284** may be positioned towards (or relatively closer to) the secondary axial end **254** of the housing **234**.

[0034] The plunger **238** may be disposed within the chamber **258** to slide or move (e.g., back and forth between the primary axial end **250** and the secondary axial end **254**) within the chamber **258**. In so doing, the plunger **238** may move or slide between an actuated state and a de-actuated state. The actuated state means a position attained by the plunger **238** when the plunger **238** is moved away from the first portion **268** of the wall **264**, towards the secondary axial end **254** of the housing **234**. In the actuated state, the plunger **238** opens the set of ports **280** and the set of conduits **284** to the chamber **258** (or to corresponding sections of the chamber **258**). The de-actuated state means a position attained by the plunger **238** when the plunger **238** is moved all the way to abut or contact the first portion **268** of the wall **264** disposed at the primary axial end **250** of the housing **234**. In the de-actuated state, the plunger **238** closes the set of ports **280** and the set of conduits **284** to the chamber **258**.

[0035] In further detail, the plunger **238** may include a spool **304**. The spool **304** may include a first piston portion **308** at a first axial end **312** of the spool **304**, a second piston portion **316** at a second axial end **320** of the spool **304**, and a stem portion **324** connecting the first piston portion **308** to the second piston portion **316**. The first piston portion **308** and the second piston portion **316** may include similar or same diameters (or cross-sectional areas) and each of which may be equal to or slightly less than a diameter (or a cross-sectional area) defined by the chamber **258**. In that manner, the plunger **238** may be accommodated within the chamber **258** so as to slide within the chamber **258** with a minimum clearance to the chamber **258**. Further, the stem portion **324** may include a diameter (or a cross-sectional area) smaller than the diameters (or cross-sectional areas) of the first

piston portion **308** and the second piston portion **316** such that an annular recess **328** may be defined around the stem portion **324**.

[0036] Further, the first piston portion **308**, the second piston portion **316**, and the stem portion **324** may be integrally formed. Also, the first piston portion **308**, the second piston portion **316**, and the stem portion **324** may be defined co-axially with respect to each other. In that manner, the plunger **238** may define a plunger axis **332** passing through each of the first piston portion **308**, the second piston portion **316**, and the stem portion **324**. When the plunger **238** is disposed or assembled into the chamber **258**, the plunger axis **332** may fall in line or be co-axially aligned with the chamber axis **262**. In so doing, a movement or the slide of the plunger **238** within the chamber **258** may be facilitated (e.g., in a back and forth pattern) along the chamber axis **262**. Also, in some embodiments, the plunger axis **332** may be co-axially aligned or may be parallel to the first port axis **288**.

[0037] It may be noted that in the de-actuated state of the plunger **238**, the chamber **258** may be closed to the set of ports **280** and the set of conduits **284** by the first piston portion **308** and the second piston portion **316**, respectively. Conversely, in the actuated state, the first piston portion **308** and the second piston portion **316** may clear and open the set of ports **280** and the set of conduits **284** to the chamber **258** (or to corresponding sections of the chamber **258**), respectively.

[0038] The plug **246** may be coupled to the housing **234** at the secondary axial end **254** of the housing **234**. As an example, the plug **246** may be inserted (e.g., by press-fitting) into the chamber **258** from the opening **276** formed at the secondary axial end **254**. In this regard, the plug **246** may define a stub portion **336** that may interfere with the chamber **258** at the opening **276** of the chamber **258** defined at the secondary axial end **254** such that the interference may help attain an interference fit of the stub portion **336** (and thus the plug **246**) with the opening **276** at the secondary axial end **254**. In that manner, a coupling between the plug **246** and the housing **234** may be attained. Other manner coupling or connection, such as a threadable coupling of the plug **246** with the opening **276** of the chamber **258** may be contemplated.

[0039] The biasing member **242** may be seated between the plunger **238** and the plug **246**. The biasing member **242** may bias the plunger **238** to the de-actuated state (or to a default state or a home state) when the plug **246** is coupled to the housing **234** at the secondary axial end **254** of the housing **234**. In some embodiments, the biasing member **242** includes a spring **340**.

[0040] The actuated state of the plunger **238** (see FIG. 5) may be attained when a lubricant supply enters into the chamber **258** through the first port **280** and a lubricant supply pressure of the lubricant supply impinges on the plunger **238** (or the first piston portion **308** of the plunger **238**) and pushes and moves the plunger **238** against a force (e.g., a spring force) of the biasing member **242** such that the plunger **238** moves in a direction towards the secondary axial end **254** to the actuated state. In the actuated state, the plunger **238** may open the first port **280'** and the second port **280''** for communication with a section (e.g., a first section **344**) of the chamber **258**, and also may open the first conduit **284'** and the second conduit **284''** for fluid communication with another section (e.g., a second section **348**) the chamber **258**. In other words, in the actuated state, the plunger **238** may divide the chamber **258** into the first section **344** and the second section **348** (see FIG. 5), with the first section **344** being defined relatively closer to the primary axial end **250** of the housing **234** and the second section **348** being defined relatively closer to the secondary axial end **254** of the housing **234**.

[0041] As may be visualized from FIG. 5, in the actuated state of the plunger **238**, the first section **344** may be defined or delimited by the housing **234** and the first piston portion **308**, or, in other words, by the first portion **268** of the wall **264**, a region of the second portion **272** of the wall **264**, and the first piston portion **308** of the plunger **238**. In that manner, in the actuated state, the set of ports **280** (e.g., the first port **280'** and the second port **280''**) may be fluidly coupled to each other through the first section **344** of the chamber **258** to define a first passageway **352** of the housing **234**. Also, in the actuated state of the plunger **238**, the second section **348** may be defined or

delimited by the housing **234**, the first piston portion **308**, and the second piston portion **316**, or, in other words, by another region of the second portion **272** of the wall **264**, the first piston portion **308**, and the second piston portion **316** of the plunger **238**. In that manner, in the actuated state, the set of conduits **284** (e.g., the first conduit **284'** and the second conduit **284''**) may be fluidly coupled to each other through the second section **348** of the chamber **258** to define a second passageway **356** of the housing **234**. It may be noted that the second section **348** and the annular recess **328** may coincide with each other.

[0042] The de-actuated state of the plunger **238** (see FIG. 4) may be attained when the lubricant supply is halted through the first port **280'** and the corresponding lubricant supply pressure is reduced. In so doing, the plunger **238** may return to the default state or the de-actuated state by having the biasing member **242** push the plunger **238** towards the primary axial end **250** of the housing **234** and, in turn, moving the plunger **238** all the way to abut or contact the first portion **268** of the wall **264** disposed at the primary axial end **250** of the housing **234**. As shown in FIG. 4, in the de-actuated state of the plunger **238**, the first piston portion **308** blocks the first passageway **352** and the second piston portion **316** blocks the second passageway **356**.

INDUSTRIAL APPLICABILITY

[0043] A method for facilitating a lubricant transfer with respect to the first bearing system **172** of the hitch system **144** is discussed. During a lubricant transfer operation or a lubricant replenishment of the first bearing system **172**, an operator may provide the valve **230** and may bring forth the valve **230** towards the hitch system **144** or the first bearing system **172** of the hitch system **144** and may fluidly couple the first passageway **352** to the lubricant supply line **198**. Thereafter, the operator may also fluidly couple the second passageway **356** to the second gallery **158**. Because the second gallery **158** facilitates a purging or a release of the lubricant (e.g., a used lubricant) from the hitch system **144** or the first bearing system **172** of the hitch system **144**, the second gallery **158** may also be interchangeably referred to as a lubricant release line **162**. In some embodiment, the fluid coupling between the first passageway **352** and the lubricant supply line **198** may be accomplished by way of a first hose **360** and the fluid coupling between the second passageway **356** and the lubricant release line **162** may be accomplished by way of a second hose **364**. Other ways to achieve such fluid coupling may be contemplated by those with skill in the art. Also, at this point, the plunger **238** of the valve **230** may be in the de-actuated state.

[0044] In some embodiments, and as shown in FIG. 3, the first hose **360** and the second hose **364** may be omitted. In such a case, the valve **230** or the housing **234** of the valve **230** may be brought closer to (e.g., to abut with) the forward frame portion **136** such that the first passageway **352** (or the second port **280''** of the first passageway **352**) may be directly and fluidly coupled with the lubricant supply line **198**, e.g., by use of a suitable coupler. Similarly, the second passageway **356** (or the first conduit **284'** of the second passageway **356**) may be directly and fluidly coupled with the lubricant release line **162**, e.g., by use of another suitable coupler.

[0045] Once the fluid coupling between the first passageway **352** and the lubricant supply line **198** and the fluid coupling between the second passageway **356** and the lubricant release line **162** is accomplished, the operator may initiate a supply of a lubricant (e.g., a new lubricant) (having a corresponding lubricant supply pressure) through the first passageway **352**, e.g., sequentially through the first port **280'**, the first section **344** of the chamber **258**, and the second port **280''** (see direction, A, FIGS. 2, 3, and 5), such that the lubricant (e.g., the new lubricant) passes through the first passageway **352** and enters into the first interface **200** defined between the first inner race **188** and the first outer race **192** of the first bearing system **172**.

[0046] More specifically, as the lubricant (e.g., new lubricant) enters into the first port **280'**, the lubricant supply pressure of the inflowing, new lubricant pushes against the plunger **238** or the first piston portion **308** of the plunger **238** such that the lubricant supply pressure overcomes the force of the biasing member **242** and causes the plunger **238** to move towards the secondary axial end **254** of the housing **234** (see direction, M, FIG. 4), thereby moving the plunger **238** to the actuated

state from the de-actuated state. In process of the movement of the plunger **238** towards the secondary axial end **254** or to the actuated state, the plunger **238** opens the first section **344** of the chamber **258** to each of the first port **280'** and the second port **280''**, thereby granting access to the inflowing, new lubricant and the associated lubricant supply pressure to be introduced into the first section **344** of the chamber **258** and the second port **280''** of the first passageway **352**. Also, in the actuated state, the plunger **238** opens both the first conduit **284'** and the second conduit **284''** to the chamber **258** (or to the second section **348** of the chamber **258**), and thus also opens the second passageway **356**.

[0047] A continued lubricant supply pressure may cause the lubricant (e.g., the new lubricant) to travel further through the first passageway **352** and exit the first passageway **352** through the second port **280''**. In that manner, the lubricant supply pressure is released out from the first section **344** of the chamber **258** and the lubricant (e.g., the new lubricant) is then delivered at or into the first interface **200**. The lubricant (e.g., the new lubricant) then exits from the first interface **200** into the lubricant case **180** (e.g., under the action of gravity). At the same time, some volume of the lubricant (e.g., the new lubricant) may also exit through the first interface **200** into the internal seal chamber **220**. Once the lubricant case **180** is filled with the lubricant (e.g., the new lubricant), a continued lubricant supply pressure may force the lubricant filled within the lubricant case **180** to further move or drain out into the duct **166** through the outlet **214** of the lubricant case **180**, and then advance or progress further into the internal seal chamber **220** so as to appropriately fill the internal seal chamber **220**.

[0048] As long as the lubricant pressure supply may be present or available, and which may be high enough to overcome the force (e.g., spring force) of the biasing member **242**, the plunger **238** may remain in the actuated state. Once the internal seal chamber **220** is appropriately filled, the operator may stop the lubricant (e.g., new lubricant) supply, thus causing a drop in the lubricant supply pressure. The drop in the lubricant supply pressure may cause the force of the biasing member **242** to overcome the lubricant supply pressure and move or return the plunger **238** to the de-actuated state in which the plunger **238** blocks the first passageway **352** and the second passageway **356**.

[0049] If an old or a used lubricant were housed within (or around) the first bearing system **172**, an introduction of the lubricant (e.g., the new lubricant), in the manner as discussed above, may simultaneously or concomitantly pressurize and/or push the used lubricant out through the lubricant release line **162** with a relatively low back pressure. In such a case, because the plunger **238** is in the actuated state, the opened, second passageway **356** may provide an exit route for the used lubricant to be purged out from the lubricant release line **162** and thus out from the hitch system **144** or from the first bearing system **172** of the hitch system **144** (see direction, B, FIGS. 2, 3, and 5).

[0050] For example, in the actuated state, the used lubricant enters into the second section **348** of the chamber **258** through the first conduit **284'**, passes annularly around the stem portion **324**, and exits out from the second section **348** of the chamber **258** (and the first bearing system **172**) through the second conduit **284''**. Effectively, in the actuated state, the valve **230** facilitates an inflow of a lubricant (e.g., a new lubricant) into the first bearing system **172** of the hitch system **144** through the first passageway **352** to cause a concomitant purge and release of a lubricant (e.g., a used lubricant) from the first bearing system **172** of the hitch system **144** through the second passageway **356**, in turn efficiently replenishing the lubricant in the first bearing system **172** of the hitch system **144**.

[0051] With the use of the valve **230**, a transfer of lubrication (i.e., both a supply of a new lubricant and a purge of a used lubricant) is efficiently and relatively easily attained. Further, because the used lubricant can be retrieved (e.g., with a low back pressure) by a use of the valve **230**, the used lubricant may also be easily collected at an exit of the second passageway **356**, e.g., into a container, such as container **368** (FIGS. 2 and 3), avoiding spillage and contamination of an outside environment. As may be understood from the description above, the valve **230** also eases and

simplifies the transfer (and/or a replenishment) of the lubricant (e.g., the new lubricant) into sealed systems, such as the first bearing system **172** of the hitch system **144**. A similar manner of lubricant replenishment or lubricant transfer operation may be contemplated for other sub-systems, such as the second bearing system **176**, of the hitch system **144**, and/or for various other systems of the machine **100** or for any machine.

[0052] It will be apparent to those skilled in the art that various modifications and variations can be made to the system of the present disclosure without departing from the scope of the disclosure. Other embodiments will be apparent to those skilled in the art from consideration of the specification and practice of the system disclosed herein. It is intended that the specification and examples be considered as exemplary only, with a true scope of the disclosure being indicated by the following claims and their equivalent.

Claims

1. A valve, comprising: a housing defining a chamber, a set of ports, and a set of conduits; and a plunger disposed within the chamber and configured to be moved to an actuated state in which the set of ports are fluidly coupled to each other through a first section of the chamber to define a first passageway and the set of conduits are fluidly coupled to each other through a second section of the chamber to define a second passageway, wherein, in the actuated state, the valve facilitates an inflow of a new lubricant into a system through the first passageway to cause a concomitant purge and release of a used lubricant from the system through the second passageway.
 2. The valve of claim 1, wherein the plunger is moved to the actuated state by a lubricant supply pressure of the new lubricant.
 3. The valve of claim 2, wherein the plunger defines a plunger axis and the set of ports includes a first port to introduce the lubricant supply pressure into the first section of the chamber and a second port to release the lubricant supply pressure out from the first section of the chamber, the first port defining a first port axis, the first port axis being defined along or parallel to the plunger axis.
 4. The valve of claim 1, wherein the plunger is configured to be moved to a de-actuated state to block the first passageway and the second passageway in the de-actuated state.
 5. The valve of claim 4 further including a biasing member to bias the plunger to the de-actuated state.
 6. The valve of claim 5, wherein the chamber is open to an end of the housing, the valve further includes a plug inserted into the chamber from the end, wherein the biasing member is seated between the plunger and the plug.
 7. The valve of claim 5, wherein the biasing member includes a spring.
 8. The valve of claim 4, wherein the plunger includes a spool with a first piston portion at a first axial end of the spool, a second piston portion at a second axial end of the spool, and a stem portion connecting the first piston portion to the second piston portion.
 9. The valve of claim 8, wherein, in the actuated state of the plunger, the first section of the chamber is delimited by the housing and the first piston portion, and the second section of the chamber is delimited by the first piston portion, the second piston portion, and the housing, and in the de-actuated state of the plunger, the first piston portion blocks the first passageway and the second piston portion blocks the second passageway.
 10. The valve of claim 8, wherein the set of conduits include a first conduit and a second conduit, and in the actuated state, the used lubricant enters into the second section of the chamber through the first conduit, passes annularly around the stem portion, and exits out from the second section of the chamber through the second conduit.
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